8.5.15(E).

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
April - May 2015

Max. Marks: 100
Class: TE
Name of the Course: Software Engineering
Course Code: UCEC602

Duration: 3hrs
Semester: VI
Branch: COMP

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q1	(a) What are the advantages of Software process model? (b) Explain phases of SDLC. (c)Explain Agile process with its advantages, explain any one agile process model?	5+5+10
Q2	(a) The values of size in KLOC and different cost drivers for a project are given below. Size=200KLOC Cost driver : s/w reliability=1.15 use of s/w tools =0.91 execution time constraint=1.00 Calculate the effort for three types of projects viz organic, semidetached and embedded, using COCOMO model (b) Explain how Gnatt chart can be used for planning and controlling small projects with suitable example? What are the limitations of gnatt chart? OR (a) Consider a project with the following data. No. of external inputs with the low complexity=10 No. of external inputs with high complexity=10 No. of external outputs with average complexity=15 No. of external inquires with average complexity=13 No. of internal logical files with high complexity=2 No. of internal logical files with low complexity=2 No. of external interface files with average complexity=7 The system has a very high transaction rate and supports several multiple communication protocols. Calculate the unadjusted as well as adjusted function points. (b) Explain Risk identification, risk projection, RMMM plan in detail	10+10
Q3	a) What is s/w quality assurance? Explain different quality matrices?	10+10

	b) Explain the difference between white box and black box testing OR a) What is the need of s/w maintenance? Explain types of s/w maintenance? b) What do you mean by s/w design, what are the characteristics of a good design ?	
Q4	a) Define software configuration management, explain change control management and version control management (b)Describe the activities done during FTR , configuration audit and status reporting .	10+10
Q5	a) Define the term web design what should be the design goals of a web based application. b) Explain security engineering	10+10